PROJECT RESULTS

Physics-Aware ML for Bio-Oil Pyrolysis

Accurate, physics-consistent prediction models and a working Pyrolysis Advisor for higher yield and lower oxygen

TEAM 2

Mohalraj Shivanethran (AIM25020)
Pranesh Subramanian (AIM25030)
Srinath Balakrishnan (AIM25040)
Umasuthan Palaniappan (AIM25041)

MENTOR

Dr. Akhil Mohan
Assistant Professor, School of Artificial Intelligence

OUTCOMES AT A GLANCE

Three things this project delivers

01

Accurate prediction

O/C model reaches R^2 0.90 on the standard literature test, matching Zhang et al. (0.895). Yield models reach R^2 up to 0.69.

02

Physics-aware models

Final yield and O/C models built on kinetics-based features and physical constraints that behave like real pyrolysis.

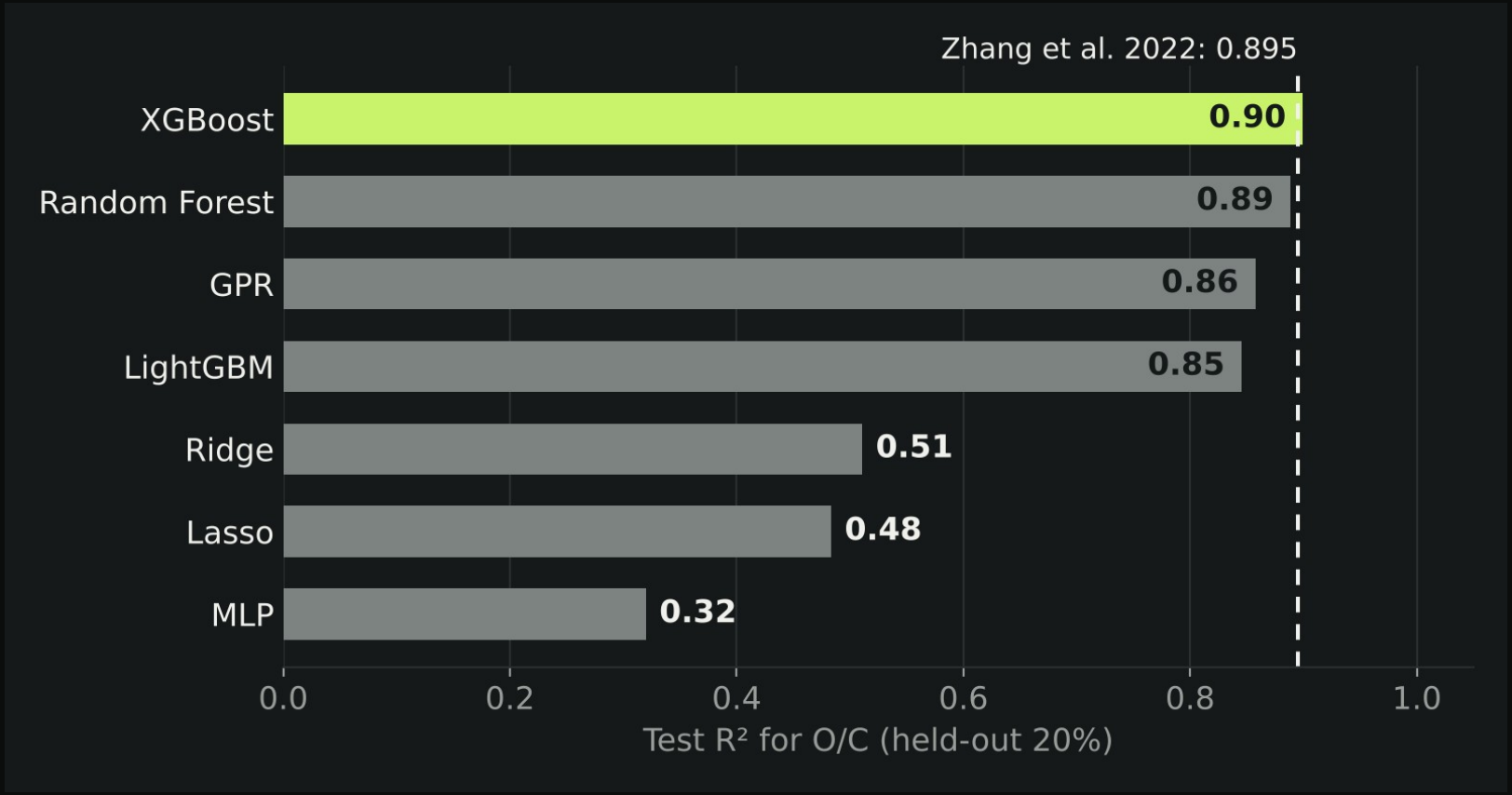
03

A working Advisor

A web tool that recommends temperature and heating rate and shows the yield/oxygen trade-off.

Data: Zhang et al. 2022 (O/C, 122 biomasses) and Pascarella et al. 2025 (yield, ~160 studies)

Our O/C model matches the published benchmark



R² 0.90

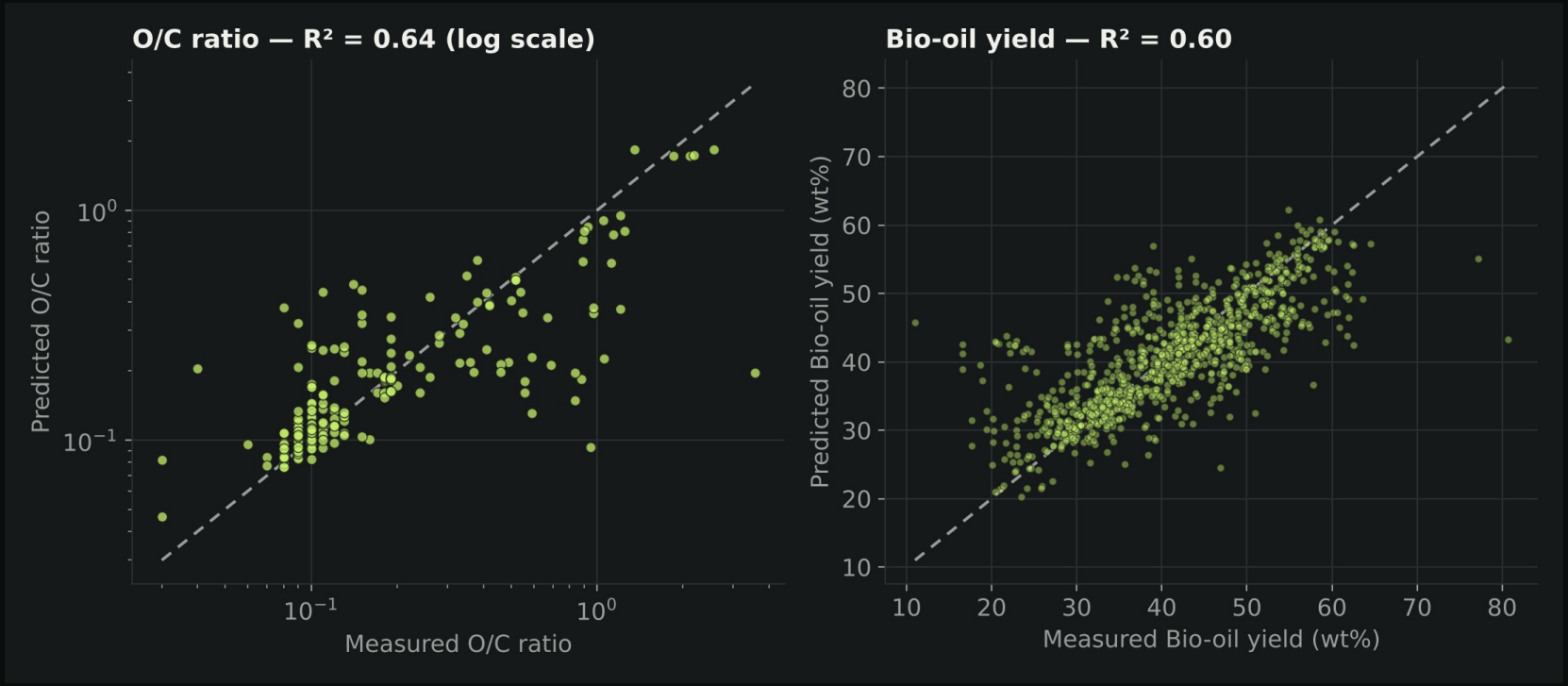
XGBoost O/C on the held-out 20% test set

0.895

Zhang et al. (2022) published O/C result

7 models compared · 232 samples · physics-informed features

Predicted vs measured for the final models



R² 0.61

O/C (log scale), mean of 10 random splits; best split 0.85; typical error 18%

R² 0.61

Yield, mean of 10 random splits; best 0.67; error 4.2 wt%

Yield model (random split)	R²	MAE
Random Forest	**0.69**	3.8 wt%
XGBoost	**0.66**	4.2 wt%
GPR	**0.60**	4.2 wt%

RESULT 3 · PHYSICS FEATURES

Physics-informed features improve every model

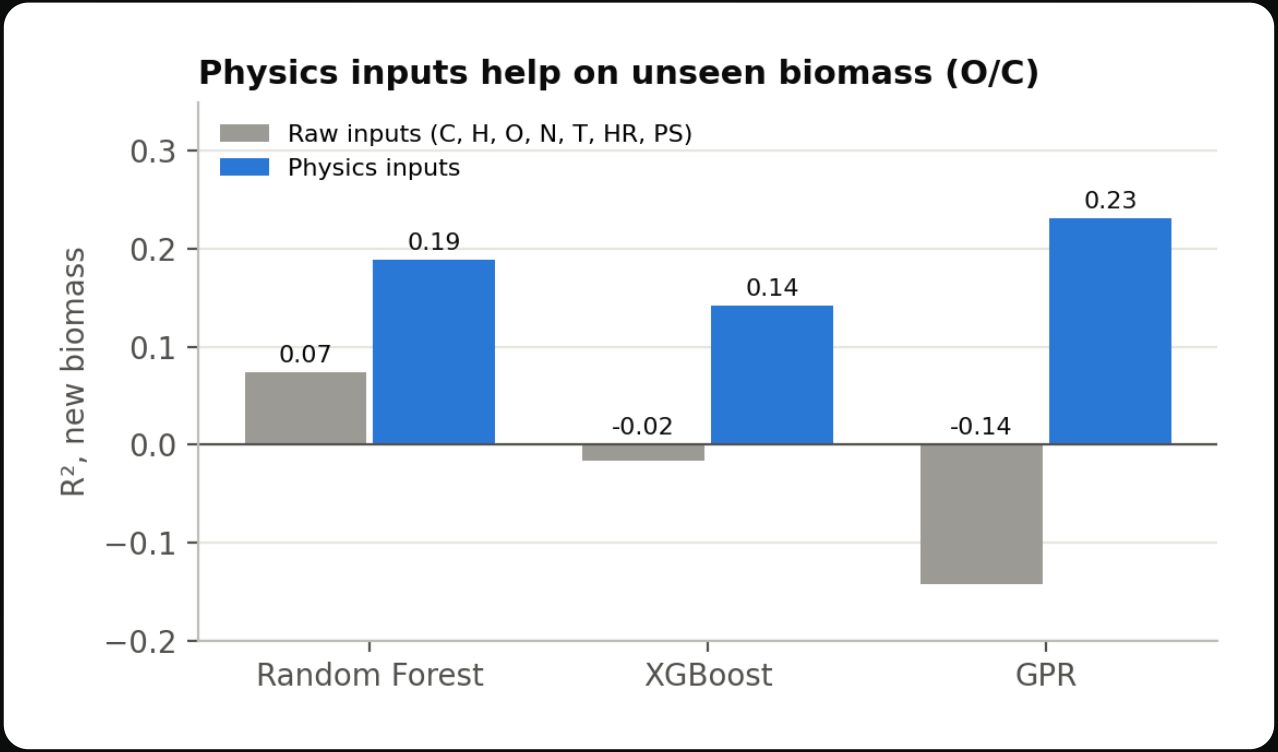


Figure: Test 2 (unseen biomass), raw vs physics inputs

0.43 → 0.63

GPR O/C R^2, Test 1: raw → physics inputs

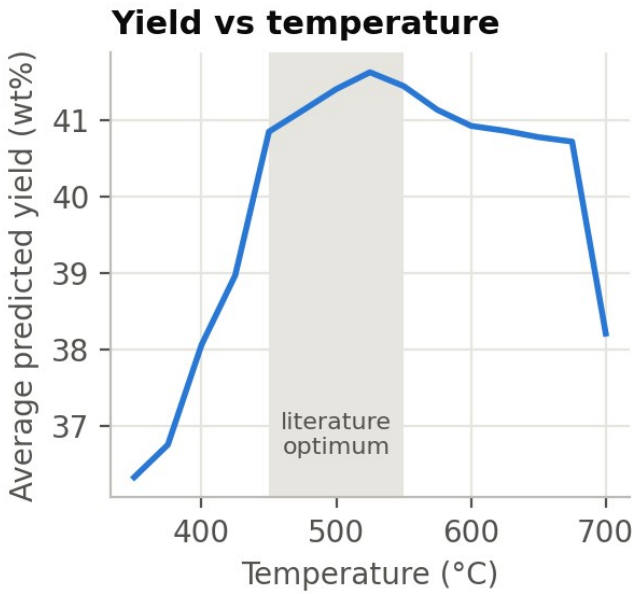
−0.14 → 0.23

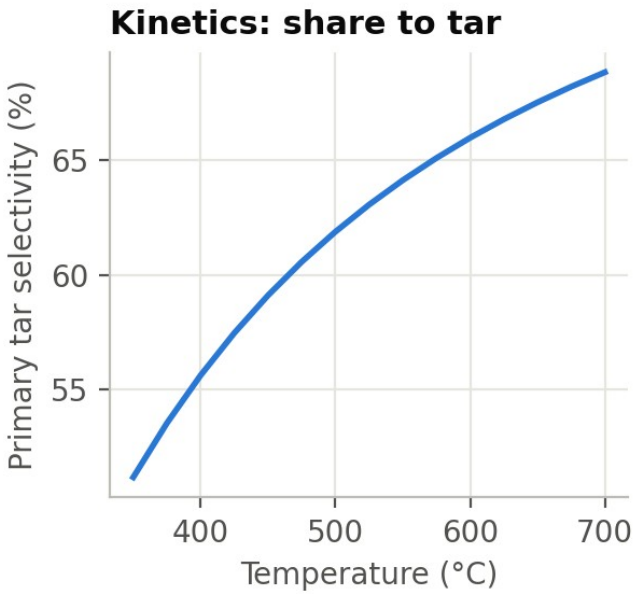
Same model, Test 2 (unseen biomass)

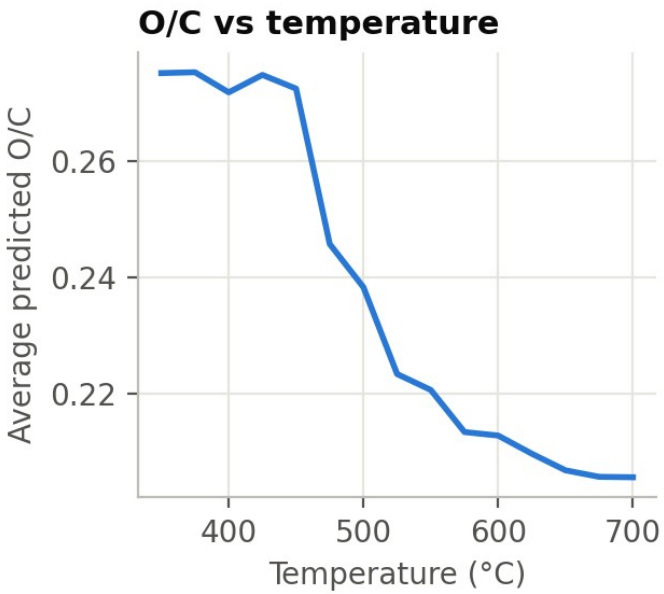
- Chan et al. (1985) Arrhenius kinetics: wood → gas / tar / char
- Kinetic tar share and pyrolysis number at each temperature
- Heating-regime and thermal-severity indices
- Molar O/C, H/C and effective H/C of the feedstock

RESULT 4 · PHYSICS CONSISTENCY

The final models behave like real pyrolysis







525°C

Yield peak (41.6 wt%) — inside the literature 450–550°C optimum

51% → 69%

Tar selectivity rises 350→700°C, explaining the yield drop from vapour cracking

0.28 → 0.21

O/C falls with temperature — stronger deoxygenation, never violated

Second test: a biomass the model has never seen

0.23

O/C R^2 when whole biomasses are held out (GPR)

0.66 → −0.18

Yield R^2, Test 1 → Test 2 (XGBoost)

Most published studies stop at Test 1. Holding out whole biomasses is harder, and the drop tells us when to calibrate with real experiments.

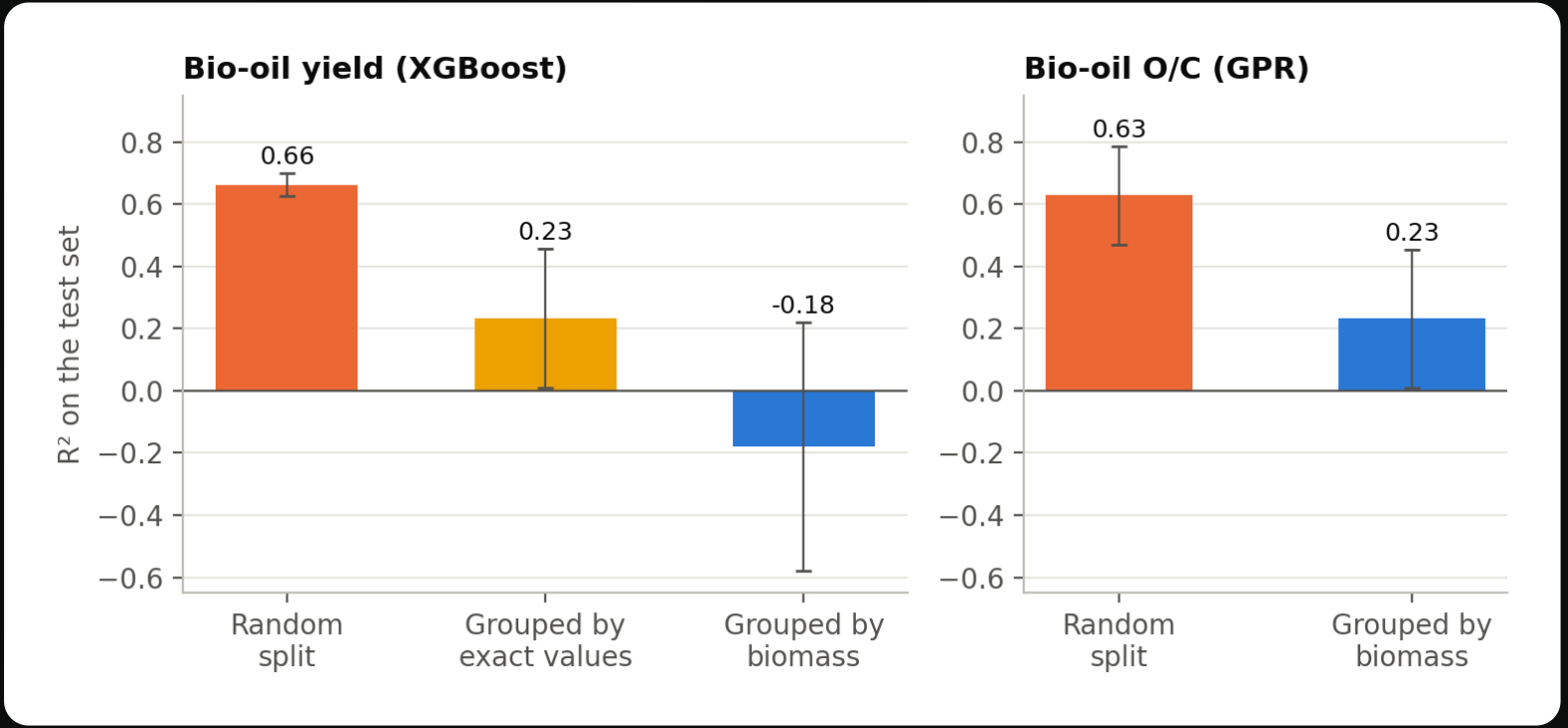
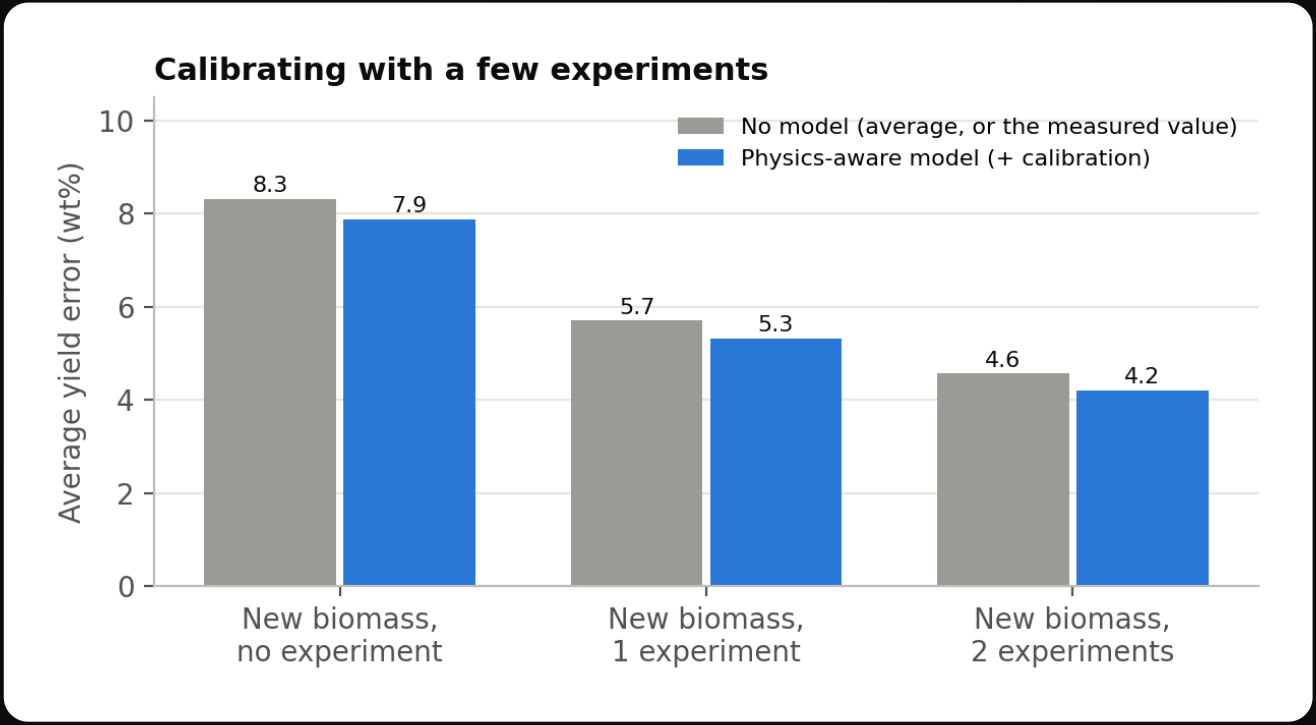


Figure: test R^2 for Test 1 (random split) vs Test 2 (grouped by biomass)

One or two real experiments sharply cut the error



~8 → 5.3 → 4.2

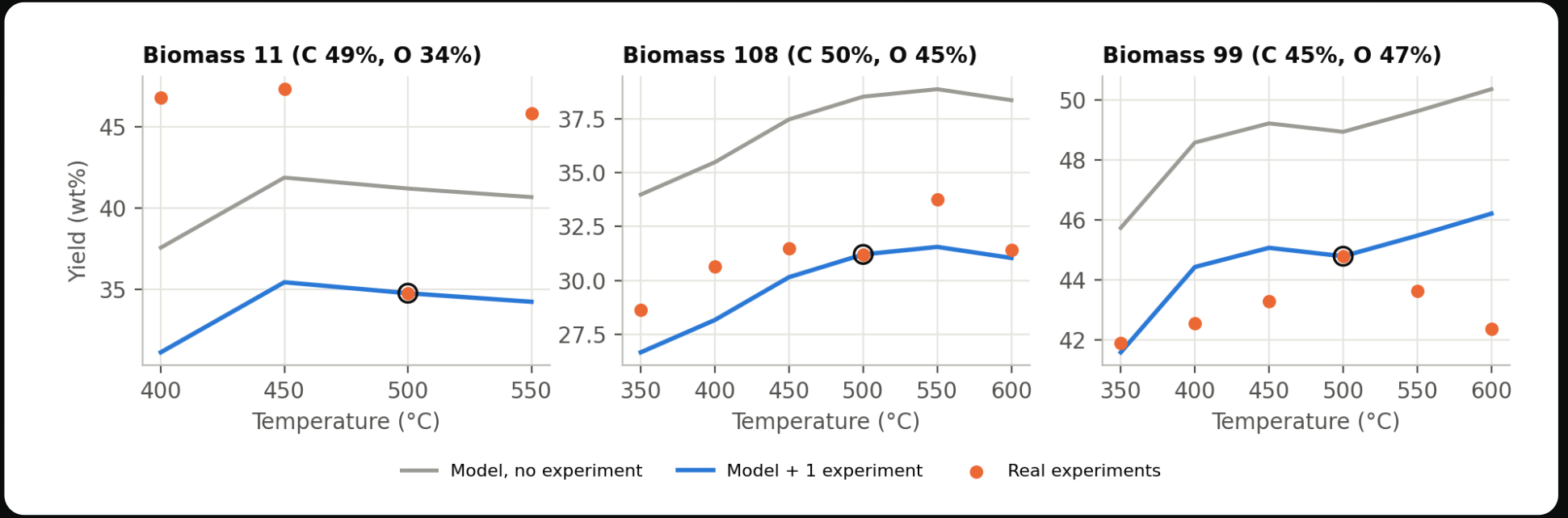
Average yield error (wt%) with 0, 1, 2 calibration experiments

±12.6 wt%

80% interval for a brand-new biomass (Test 2) with no experiments

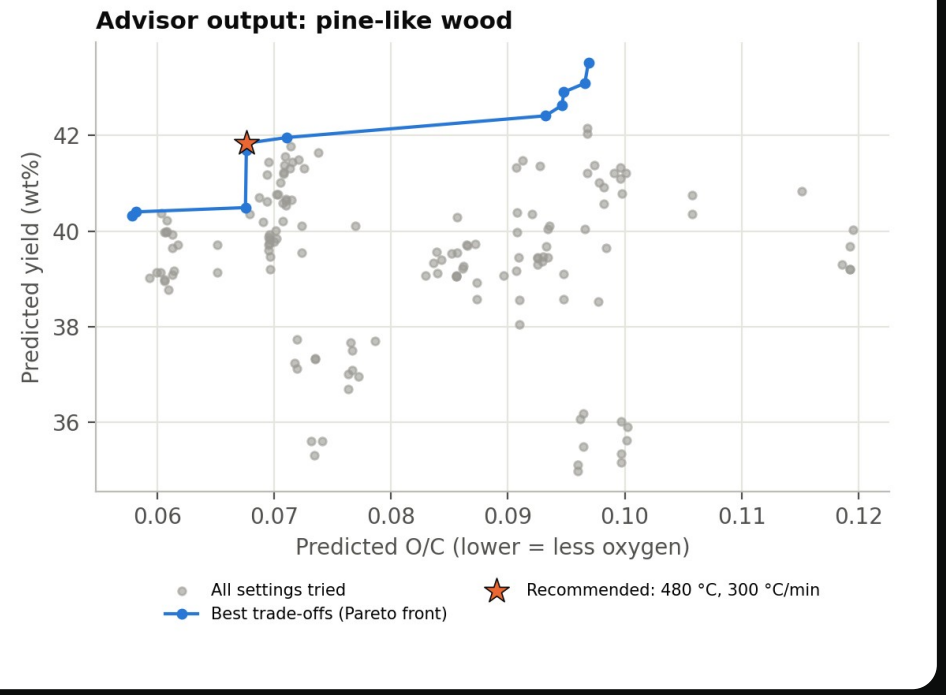
Calibrated model beats simply reusing the measured value (5.7, 4.6 wt%) because it adds the learned response to conditions.

Correct optimum temperature for 2 of 3 unseen biomasses



Biomass	Real best T	Model best T	Error, 0 exp.	Error, 1 exp.
11 (C 49%, O 34%)	450°C	450°C ✓	6.6 wt%	13.0 wt%
108 (C 50%, O 45%)	550°C	550°C ✓	5.6 wt%	1.7 wt%
99 (C 45%, O 47%)	500°C	600°C	6.0 wt%	1.9 wt%

Recommendations that show the trade-off — and know their limits

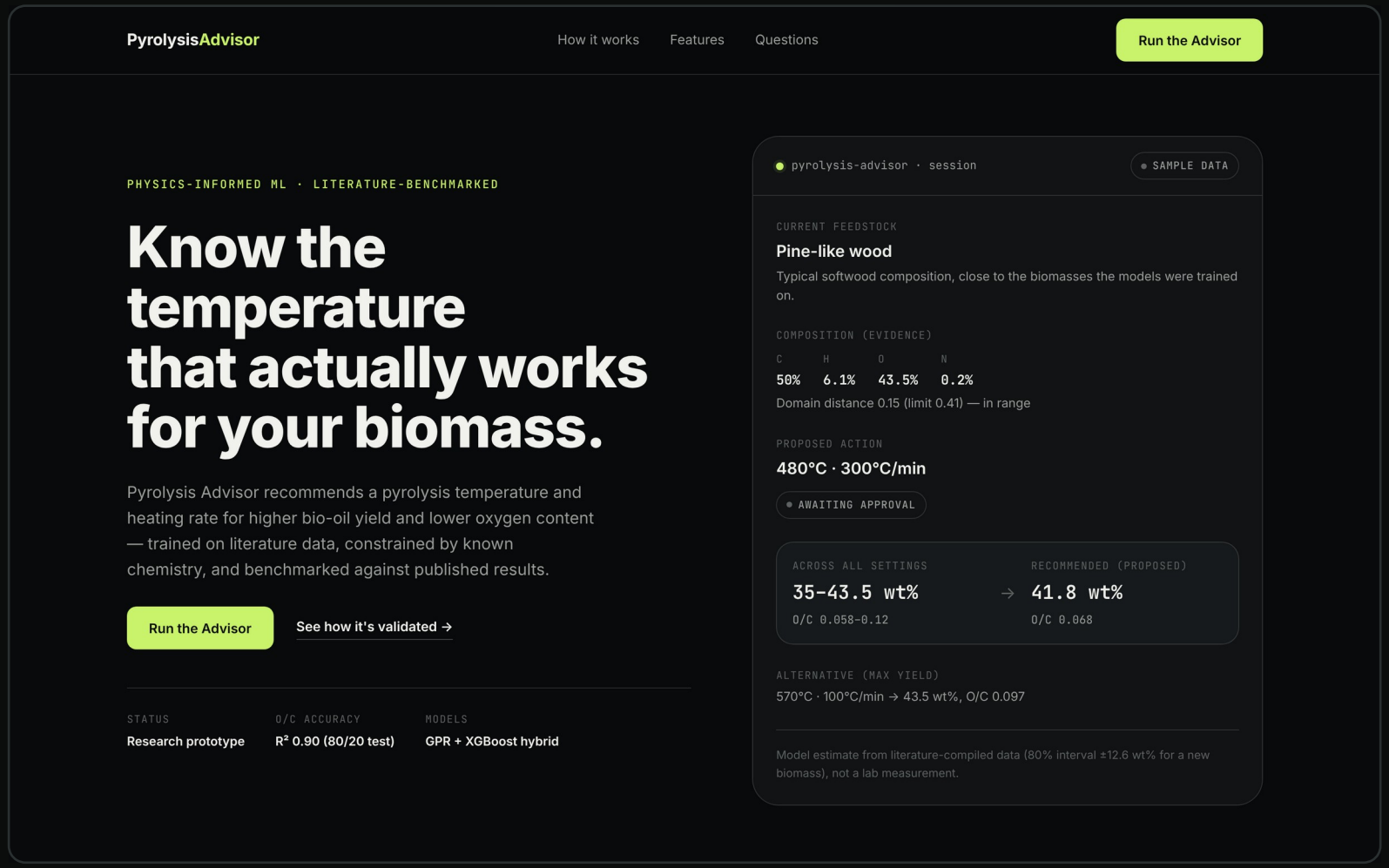


Feedstock	Status	Recommended	Max-yield alternative
Pine-like wood	In range	480°C, 300°C/min → 41.8 wt%, O/C 0.068	570°C → 43.5 wt%, O/C 0.097
Rice husk-like	In range	510°C, 10°C/min → 41.6 wt%, O/C 0.062	570°C → 44.7 wt%, O/C 0.113
Microalgae-like (N 8.5%)	Flagged	Withheld — distance 0.52 > 0.41	Needs 1–2 experiments
Coal-like (C 80%)	Flagged	Withheld — distance 1.20 > 0.41	Not biomass

For pine, the last 1.7 wt% of yield costs about 40% more oxygen in the oil.

WORKING MODEL · LIVE DEMO

The Advisor as a web app

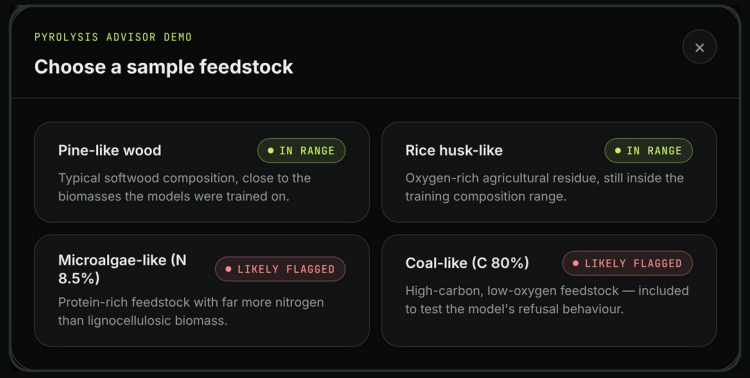


- React + TypeScript web app
- Pick a feedstock, review the recommended conditions, then accept or reject
- Every screen states it uses sample results from our validation

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Run locally: advisor-frontend → npm run dev
```

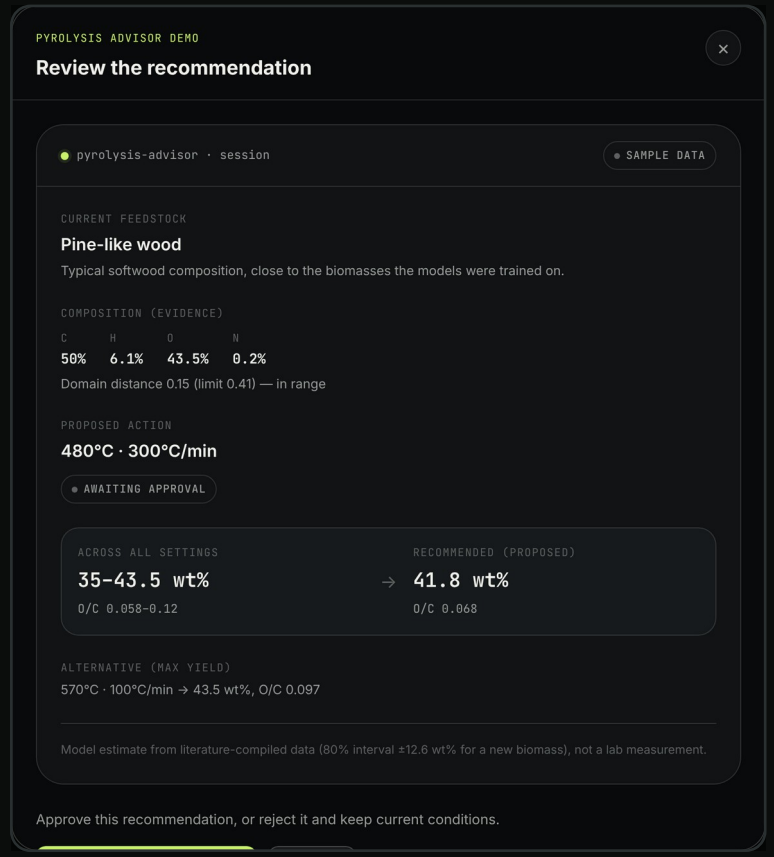
Choose → review → decide

1 Choose a sample feedstock

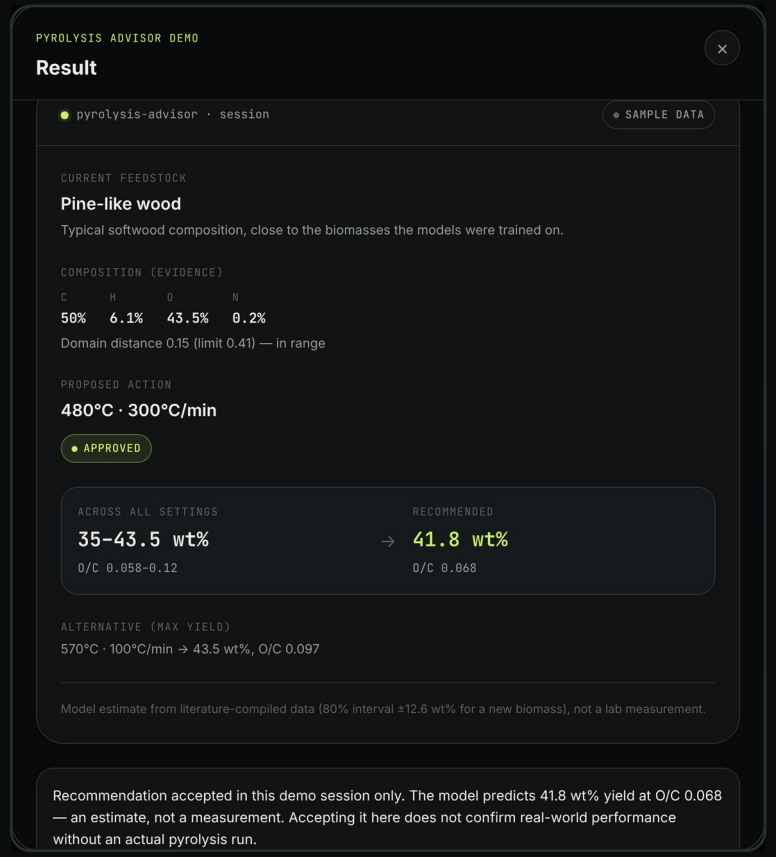


Two feedstocks are in range; two are flagged before any number is shown.

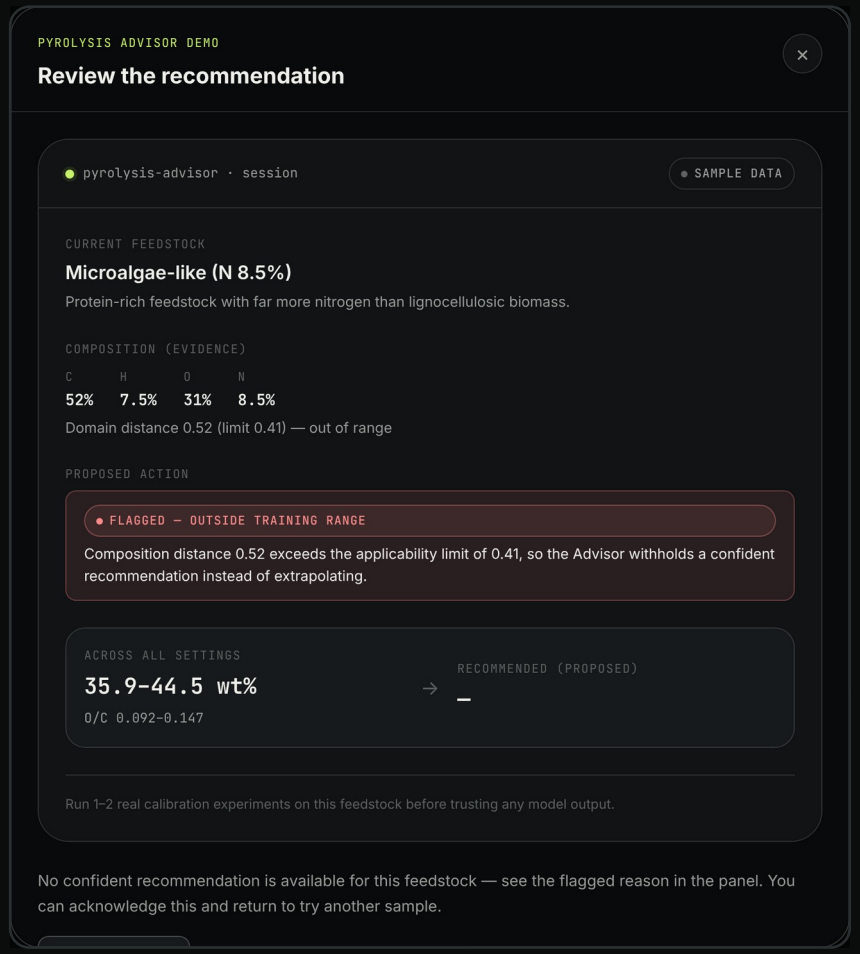
2 Review recommendation and range



3 Accept — result shown honestly



It refuses to guess on unfamiliar feedstock



0.52 > 0.41

Microalgae-like feedstock: composition distance exceeds the applicability limit

- No recommendation is shown, only the reason
- User is told to run 1–2 calibration experiments first
- Coal-like input (distance 1.20) is refused the same way

SUMMARY

Results scorecard

Metric	Test 1: standard 80/20 split	Test 2: unseen biomass
O/C R^2, best benchmark model (XGBoost)	**0.90**	—
O/C R^2, final Advisor model (log scale)	**0.61**	**0.27**
Yield R^2, best single model (Random Forest)	**0.69**	**−0.25**
Yield R^2, final Advisor model	**0.61**	**0.07**
Yield error after 1 / 2 calibration experiments	—	**5.3 / 4.2 wt%**
Correct optimum temperature, 3 held-out biomasses	—	**2 of 3**
Out-of-range feedstocks flagged	—	**2 of 2**

What we would do next

LIMITATIONS

- Literature data mixes labs; residence time rarely reported
- Yield and O/C come from two separate datasets
- Wide uncertainty for a brand-new biomass
- Lower accuracy on brand-new biomass (Test 2)
- Web demo runs on pre-computed sample results

NEXT STEPS

- Leave-one-study-out testing to isolate lab effects
- Model organic (water-free) yield on 413 rows
- Collect consistent single-reactor data and retrain
- Connect the web app to the trained models via API

Thank you

Questions and live demo

github.com/umasuthanpalaniappan/BioOil-Automation